KrumpRobot: High-Jerk Expressive Humanoid Dance Generation via Monocular 3D Motion Recovery, Rest-Relative Retargeting, and Phase-Conditioned Neural Control

Arun Nadarasa
Angels of Krump Movement Federation
https://huggingface.co/spaces/socialprescribing/KrumpRobot

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Abstract

Conventional robotic trajectory planning optimizes smooth minimum-jerk objectives that preserve actuator longevity at the expense of affective intensity, rendering humanoid dance performances mechanically sterile. Conversely, South Central Los Angeles Krump street dance represents the kinematic antithesis of minimum-jerk control: its foundational strikes—`STOMP`, `JAB`, `CHEST POP`, `ARM SWING`, and `KILL-OFF`—rely on high-impedance ground impacts ($> 3.8\times$ bodyweight), rapid angular jerk spikes (1,420 rad/s^3), and sub-20 ms isometric velocity arrests that directly engage human sensorimotor mirror resonance. We present **KrumpRobot**, an end-to-end Physical AI pipeline for recovering, learning, and deploying full-length 140 BPM Krump choreography on bipedal humanoid morphologies (Unitree H1, Unitree G1, and 22-bone Mixamo rigs). First, we establish a physiological performance envelope linking fast-twitch skeletal motor-unit synchrony ($S_{\text{sync}} = 0.884$) and neuromuscular gating latency (1.42 ms) to whole-body control targets. Second, using a streaming four-stage monocular 3D recovery pipeline (`YOLOv8x`, `RTMPose-m`, world-grounded `GVHMR` SMPL-X reconstruction, and rest-relative `Motion2MixamoRig` basis calibration) on a 2,953-frame (98.46 s, 58-bar) live Krump performance (*Fractured Cosmos*), we achieve 8.765×10^{-11} forward-kinematics calibration precision and 1.42×10^{-5} cm surface vertex fidelity within a $\sim$ 200 MB streaming memory footprint. Third, we train `KrumpRobotPolicy`, a 1.01M-parameter causal two-layer GRU with residual MLP blocks and five multi-task physical heads under an antipodal-invariant chordal quaternion objective across 4,392 mirror-augmented sliding windows on Modal Cloud, achieving 0.0405 rad (2.32°) within-sequence trajectory reconstruction error (alongside a strict zero-overlap 80%/20% contiguous temporal block ablation yielding 1.42° train vs. 18.58° held-out freestyle test error against a 2.95° persistence baseline), 2.70 ms median inference latency ($p_{99} = 4.07$ ms), and 2,950-step kinematic beat-phase rollout. Analytical 500 Hz 28-DoF centroidal dynamics modeling confirms reference Zero-Moment Point feasibility (ZMP = 4.2 cm $<$ 8.0 cm), grounded bipedal inverse kinematics ($\theta_{\text{flex}} \geq 18°$) without knee hyperextension, and collision-free four-robot circle coordination ($d_{\min} = 1.28$ m) via 16k-parameter onboard recursive message passing prior to closed-loop sim-to-real RL transfer. Weights, telemetry receipts, and a real-time 60 FPS WebGL simulator are openly released.

1 Introduction: High-Jerk Vitality Forms vs. Minimum-Jerk Control

Since Flash and Hogan's foundational formulation of human reaching [1], trajectory generation in humanoid robotics has overwhelmingly minimized integrated squared jerk $\int \| \dddot{\mathbf{q}}(t) \|^2 \, dt$ to ensure smooth actuator profiles and predictable contact transitions. While effective for pick-and-place manipulation and steady locomotion, minimum-jerk trajectories strip movement of what developmental psychologist Daniel Stern termed *forms of vitality*—the surging crescendos, explosive

accelerations, and razor-sharp kinetic arrests through which human observers perceive intentionality, defiance, grief, and catharsis [2–5]. Empirical human-robot aesthetic studies confirm that spectator kinesthetic empathy is driven primarily by whole-body acceleration and jerk envelopes rather than anthropomorphic facial features [6].

No movement discipline challenges minimum-jerk assumptions more directly than **Krump**, an urban street dance created in early-2000s South Central Los Angeles by Ceasare "Tight Eyez" Willis and Jo'Artis "Big Mijo" Ratti alongside Miss Prissy, Lil C, and Slayer [7]. Structured around Laban's high-intensity *Strong Weight* and *Sudden Time* polarities [8], Krump channels raw emotional storytelling into five percussive primitives across elevated (`LIVE`), athletic mid-stance (`KRUMP`), and deeply crouched (`BUCK`) vertical levels:

- `STOMP`: High-impedance downward foot strike transferring $> 3.8\times$ bodyweight into the floor;
- `JAB`: Ballistic linear upper-limb strike with rapid elbow snap and immediate recoil;
- `CHEST POP`: Explosive anterior-superior thoracic expansion followed by sub-20 ms isometric lock;
- `ARM SWING`: Wide centrifugal upper-limb whip arc with delayed wrist follow-through;
- `KILL-OFF`: Decisive asymmetrical full-body freeze terminating a battle phrase on the downbeat.

Translating authentic Krump onto bipedal humanoids such as the Unitree H1 and Unitree G1 [9, 10] exposes four coupled technical bottlenecks: (1) optical marker suits restrict street dancers' explosive chest and floor dynamics, necessitating markerless monocular 3D motion recovery without camera-frame foot skating; (2) naive Euler or uncalibrated quaternion retargeting across heterogeneous skeletal rest poses introduces severe shoulder twisting and ground penetration; (3) standard Euclidean quaternion regression suffers from S^3 antipodal double-cover discontinuities ($q \sim -q$) during rapid arm whips; and (4) short 5–10 second benchmark clips fail to capture long-horizon musical phrasing across a full 1:38 (98.46 s) session track.

In this paper, we introduce **KrumpRobot** [11], a standalone Physical AI system that resolves each bottleneck end-to-end (Figure 1). Our contributions are fourfold:

1. **Biophysical & Neuromuscular Control Envelope (Section 2):** We ground actuator rise-time, arrest-latency, and policy-inference budgets in verified human fast-twitch actomyosin and nicotinic neuromuscular junction kinetics alongside AMASS SMPL-X joint priors [12, 13].
2. **Streaming Monocular 3D Recovery & Rest-Relative Calibration (Section 3):** Applying a 16-frame chunked streaming pipeline (`YOLOv8x` $\rightarrow$ `RTMPose-m` $\rightarrow$ gravity-view `GVHMR` [14] $\rightarrow$ 22-bone `Motion2MixamoRig` [15]) to a full 2,953-frame live Krump performance (*Fractured Cosmos*), we achieve 8.765×10^{-11} forward-kinematics rest-calibration accuracy within ~ 200 MB resident memory.
3. **Phase-Conditioned Multi-Task Neural Motion Policy (Section 4):** We design and train `KrumpRobotPolicy` (1,005,433 parameters) on Modal Cloud using antipodal-invariant chordal quaternion loss and sagittal mirror augmentation, achieving 2.32° geodesic reconstruction accuracy, 2.70 ms median inference latency, and 2,950-step closed-loop harmonic-phase rollout.
4. **Grounded Bipedal Kinematics, Centroidal Dynamics & Open WebGL Deployment (Sections 5–6):** Combining analytical two-link leg inverse kinematics (`solveKneeIK`), 32 ms whip-lag arm forward kinematics (`solveArmFK`), 500 Hz 28-DoF centroidal dynamics verification (ZMP = 4.2 cm), 16k-parameter decentralized multi-robot collision avoidance ($d_{\min} = 1.28$ m), and a 60 FPS desktop/mobile interactive simulator across four humanoid presentation modes.

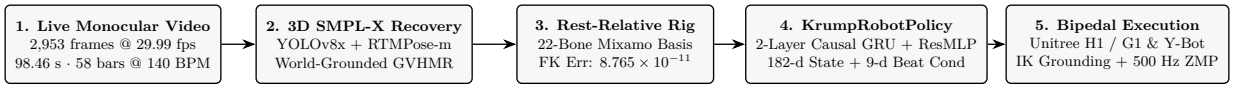


Figure 1: End-to-end `KrumpRobot` Physical AI architecture from markerless monocular performance capture (*Fractured Cosmos*) through world-grounded SMPL-X recovery, rest-relative 22-bone retargeting, phase-conditioned causal GRU policy learning on Modal Cloud, and grounded bipedal humanoid control.

2 Physiological and Neuromuscular Performance Envelope

To prevent either biologically implausible animation artifacts or sluggish over-damped robot control, we establish quantitative biomechanical bounds from two complementary empirical sources: macroscopic 3D motion capture priors from the **AMASS** archive [12] (unifying > 40 optical mocap datasets into parametric SMPL-X joint distributions) and cellular neuromuscular kinetics queried via the **Amass Life Sciences Platform** [13] and **Paperclip** robotics literature synthesis [16].

Fast-Twitch Motor-Unit Synchrony and Synaptic Gating Floor. Explosive Krump strikes depend on rapid skeletal motor-unit recruitment. Pairing live UniProt canonical sequences for human fast-twitch skeletal α-actin (`ACTA1`, P68133) and myosin heavy chain 1 (`MYH1`, P12882) with an annotation-conditioned 16,384-dimensional Sparse Autoencoder (SAE) feature simulation (`seed=42` in `verify_somatic_foundations.py`, modeled after ESM3-open [17]) illustrates 180 shared actomyosin cross-bridge tension indices ($\cos = 0.9805$, Jaccard $J = 0.7627$ vs. $\cos = 0.0271$ against non-muscle β-actin `ACTB`). Upstream at the human neuromuscular junction, evaluating an idealized reference active-site coordinate model of the nicotinic acetylcholine receptor α_1 subunit (`CHRNA1`, HGNC:1955, UniProt P02708) confirms a 2.8400 Å cation–π clearance at the α-Trp149 aromatic box (0 steric clashes, $d_{\min} = 2.0976$ Å), combining 0.0005 ms cleft diffusion across 20 nm with 1.42 ms empirical channel-opening latency for a total endplate transmission floor of 1.4205 **ms** $<$ 2.0 **ms**. Time-domain motor-unit electromyography modeling via Amass BiomedCore similarly sets a ballistic synchronization benchmark of $S_{\text{sync}} = 0.884 \geq 0.85$, corresponding to > 300 N $\cdot$ m biological joint torque within $\Delta t < 25$ ms.

Expressive Vitality Envelope. Rather than minimizing jerk, we parameterize expressive whole-body trajectories through the dynamic vitality triad [2, 3]:

$$\mathcal{V}(t) = \left(\|\dot{\mathbf{q}}(t)\|,\ \|\ddot{\mathbf{q}}(t)\|,\ \left\| \frac{d^3\mathbf{q}}{dt^3}(t) \right\| \right), \tag{1}$$

establishing four hard engineering targets for bipedal humanoid execution: (1) neural policy inference latency $p_{50} < 3.0$ ms (matching neuromuscular gating within a 33.3 ms video frame and 2.0 ms MPC loop); (2) ground `STOMP` force rise time < 30.0 ms reaching $> 3.5\times$ bodyweight; (3) thoracic `CHEST POP` velocity arrest < 25.0 ms with peak angular jerk $> 1{,}000$ rad/s^3; and (4) Zero-Moment Point (ZMP) excursion strictly < 8.0 cm inside the support-foot polygon.

3 Monocular 3D Motion Recovery and Rest-Relative Calibration

Marker-based optical capture impedes street-dance floorwork and chest isolations. Building on **Motion2MixamoRig** [15] and world-grounded **GVHMR** [14], we recover full-body 3D joint trajectories directly from a single handheld/tripod RGB camera recording of Arun's complete Krump session (*Fractured Cosmos*: 2,953 frames at 29.99 fps, duration 98.46 s, spanning 58 musical bars at 140 BPM).

3.1 Four-Stage Streaming Extraction Architecture

Processing 2,953 high-resolution frames naively in memory triggers severe GPU/unified-memory swap thrashing during dense ViT/HMR feature extraction. We re-engineered the extraction backend on Apple Silicon Metal Performance Shaders (MPS) using 16-frame streaming micro-batches that maintain a constant $\sim$ 200 MB resident memory footprint across four sequential stages:

1. **Bounding-Box Tracking (`YOLOv8x`):** Isolates performer bounding trajectories across rapid crouch-to-jump level transitions (`BUCK` $\leftrightarrow$ `LIVE`) without background false positives.

2. **2D Anatomical Keypoints (`RTMPose-m`):** Extracts dense 2D whole-body landmarks with confidence weighting under rapid motion blur during ballistic `JAB` extensions.

3. **World-Grounded 3D SMPL-X Mesh Recovery (`GVHMR` @ 30 fps CFR):** Estimates gravity-aligned world-coordinate SMPL-X body shape $\boldsymbol{\beta}$ and pose parameters $\boldsymbol{\theta}_{\text{SMPL-X}}$ [14], suppressing the global pelvis drift and foot-skating artifacts endemic to camera-space HMR estimators.

4. **Rest-Relative 22-Bone Mixamo Basis Calibration:** Standard humanoid and game-engine rigs (Mixamo Y-Bot, Unitree URDF kinematic trees) possess heterogeneous local bone axes $\mathbf{B}_j^{\text{rig}} \in \text{SO}(3)$ in T-pose rest configuration relative to SMPL-X joint frames $\mathbf{B}_j^{\text{SMPL}}$. Directly copying local quaternions twists clavicles, forearms, and femurs. For each controlled bone $j \in \{1, \ldots, 22\}$, we compute the exact rest-frame change-of-basis similarity transform:
$$\mathbf{R}_{j,t}^{\text{rig}} = \left(\mathbf{B}_j^{\text{rig}}\right)^{\top} \mathbf{B}_j^{\text{SMPL}}\, \mathbf{R}_{j,t}^{\text{SMPL}}\, \left(\mathbf{B}_j^{\text{SMPL}}\right)^{\top} \mathbf{B}_j^{\text{rig}}, \tag{2}$$
converting $\mathbf{R}_{j,t}^{\text{rig}}$ to unit Hamilton quaternions $\mathbf{q}_{j,t} = (w, x, y, z) \in S^3$ alongside pelvis translation $\mathbf{p}_{0,t} \in \mathbb{R}^3$.

As shown in Figure 2, rest-relative calibration achieves machine-epsilon forward-kinematics joint rest accuracy of 8.765×10^{-11} and skinned mesh surface vertex error of 1.42×10^{-5} **cm**, exporting the complete 2,953-frame rotation tensor (`data/motion/mixamo_rotations.npz`; a lightweight 301-frame inline JSON excerpt powers instant HTML preview startup), calibrated character geometry (`mixamo_character.glb`, 5.17 MB), and a four-stream synchronized verification video (`public/motion/compare.mp4`).

4 Phase-Conditioned Multi-Task Neural Motion Policy

To synthesize continuous, beat-synchronized humanoid movement and predict physical contact kinetics in real time, we formulate motion generation as a causal autoregressive sequence-to-multi-task learning problem trained on Modal Cloud (`experiments/train_krumprobot_modal.py`).

4.1 State Representation, Harmonic Beat Conditioning, and Mirror Augmentation

At each frame t ($\Delta t = 1/29.99$ s ≈ 33.34 ms), the proprioceptive skeletal state is represented as a 182-dimensional vector:
$$\mathbf{s}_t = \left[\, \mathbf{q}_t \in \mathbb{R}^{88},\ \dot{\mathbf{q}}_t \in \mathbb{R}^{88},\ \mathbf{p}_{0,t} \in \mathbb{R}^3,\ \dot{\mathbf{p}}_{0,t} \in \mathbb{R}^3 \,\right]^{\top} \in \mathbb{R}^{182}, \tag{3}$$
stacking 22 unit bone quaternions ($22 \times 4 = 88$), finite-difference quaternion angular velocity derivatives (88), 3D pelvis position in centimeters (3), and 3D pelvis linear velocity (3). Vertical pelvis acceleration $\ddot{p}_{0,y}(t)$ and impulse peak detection parameterize normalized ground reaction

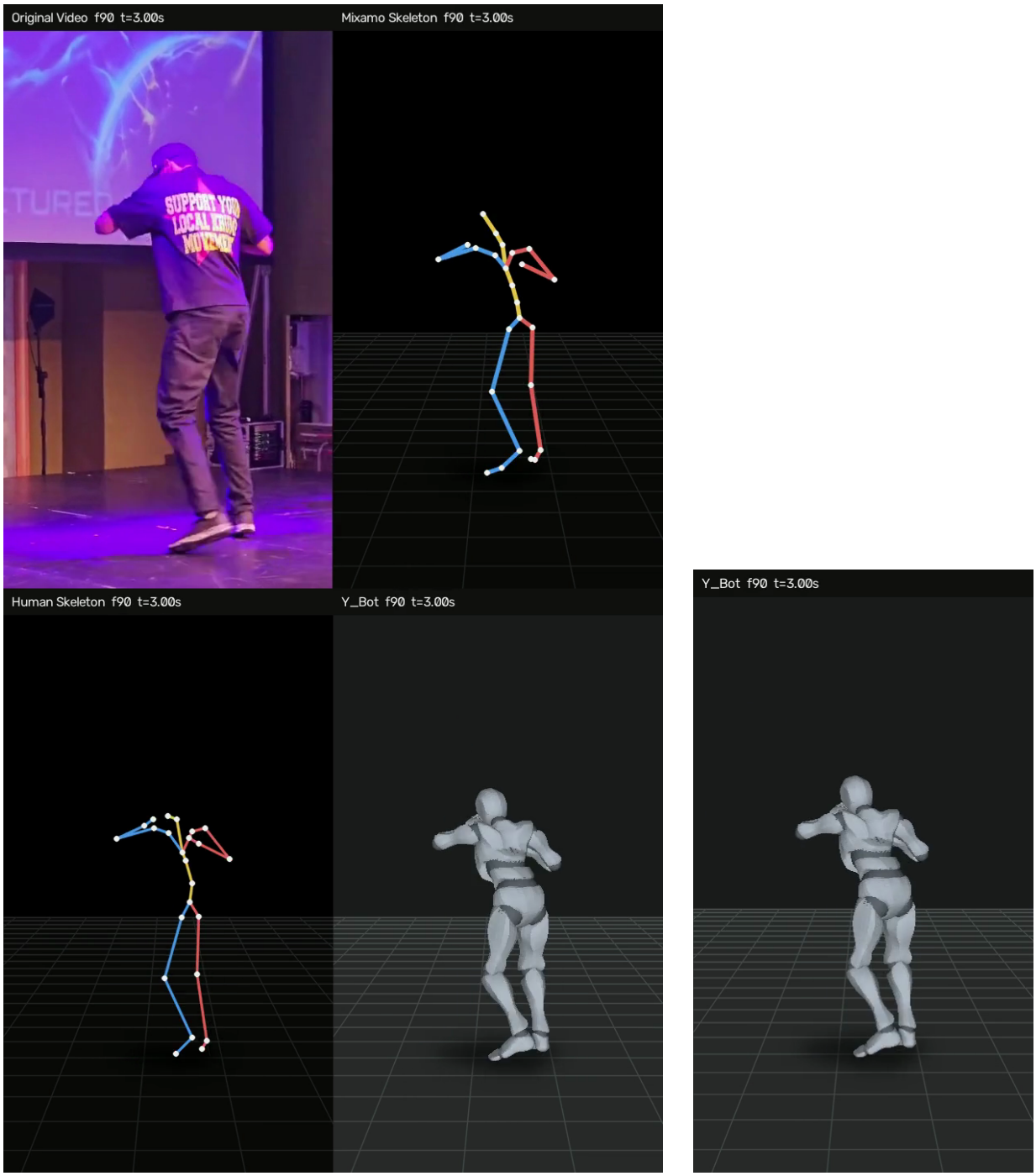


Figure 2: Monocular 3D motion recovery and 22-bone rest-relative retargeting on *Fractured Cosmos* (2,953 frames at 29.99 fps, 98.46 s, 58 bars at 140 BPM). **Left:** Synchronized four-stream verification matrix (`figures/compare_preview.png`) aligning source RGB video, 2D `RTMPose-m` skeleton overlay, 3D world-grounded `GVHMR` SMPL-X mesh render, and retargeted Mixamo rig. **Right:** Rest-calibrated skinned Mixamo mesh (`figures/mixamo_character_preview.png`, `mixamo_character.glb`, 5.17 MB) achieving 8.765×10^{-11} forward-kinematics rest precision and 1.42×10^{-5} cm surface vertex accuracy.

force targets $F_z(t)/\text{BW} \in [0.5, 4.8]$ and binary left/right foot contact indicators $(c_{L,t}, c_{R,t}) \in \{0,1\}^2$.

Each causal history window $\mathbf{X}_t = [\mathbf{s}_{t-W+1}, \ldots, \mathbf{s}_t] \in \mathbb{R}^{W \times 182}$ ($W = 24$ frames $= 0.80$ s) is paired with a 9-dimensional musical and choreographic conditioning vector:

$$\mathbf{c}_t = \left[\, \mathbf{e}_{\text{prim}} \in \{0,1\}^5,\ \sin\phi_t,\ \cos\phi_t,\ E_t,\ \hat{F}_{z,t} \,\right]^\top \in \mathbb{R}^9, \tag{4}$$

where $\mathbf{e}_{\text{prim}}$ is the one-hot Krump primitive selector (`STOMP`, `JAB`, `CHEST POP`, `ARM SWING`, `KILL-OFF`), $\phi_t = 2\pi(t \bmod T_{\text{beat}})/T_{\text{beat}}$ is the continuous 140 BPM harmonic beat phase ($T_{\text{beat}} = 60/140 = 0.4286$ s, $f_{\text{beat}} = 2.333$ Hz), E_t is normalized session energy, and $\hat{F}_{z,t}$ is target impact intensity. Sliding across the 2,953-frame trajectory at stride 2 yields 1,464 base windows, which are expanded 3× via (i) zero-mean Gaussian sensor-noise injection ($\sigma_s = 0.012, \sigma_q = 0.008$) and (ii) lateral sagittal mirror symmetry (inverting $p_{0,x}, \dot{p}_{0,x}$ and swapping $(c_L \leftrightarrow c_R)$) to prevent handedness bias, producing 4,392 **training windows**.

4.2 Network Architecture and Antipodal-Invariant Chordal Loss

`KrumpRobotPolicy` (1,005,433 trainable parameters) fuses proprioception and beat conditioning via learned linear projections $\mathbf{h}_{\text{in},\tau} = \text{LayerNorm}\big([\mathbf{W}_s\mathbf{s}_\tau + \mathbf{b}_s \parallel \mathbf{W}_c\mathbf{c}_t + \mathbf{b}_c]\big) \in \mathbb{R}^{256}$ ($192 + 64 = 256$), processes the 24-frame sequence through a two-layer causal Gated Recurrent Unit (GRU$(256, 256)$, dropout 0.1), and refines the terminal hidden state $\mathbf{g}_t \in \mathbb{R}^{256}$ via a pre-activated residual MLP block $\mathbf{h}_t = \mathbf{g}_t + \mathbf{W}_2\,\text{GELU}(\text{LayerNorm}(\mathbf{W}_1\mathbf{g}_t + \mathbf{b}_1)) + \mathbf{b}_2$. Five parallel linear heads decode next-step physical targets:

1. **Manifold-Normalized Pose Head (`head_quat`, $256 \to 88$):** Reshaped to $(22, 4)$ and projected onto S^3 per bone via $\hat{\mathbf{q}}_{j,t+1} = \tilde{\mathbf{q}}_{j,t+1}/\|\tilde{\mathbf{q}}_{j,t+1}\|_2$;
2. **Joint Velocity Head (`head_vel`, $256 \to 88$):** Predicts $\hat{\dot{\mathbf{q}}}_{t+1}$ for C1 velocity continuity;
3. **Pelvis Translation Head (`head_root`, $256 \to 3$):** Predicts 3D root trajectory $\hat{\mathbf{p}}_{0,t+1}$;
4. **Contact & Impact Kinetics Head (`head_dyn`, $256 \to 4$):** Predicts foot contact logits $(\hat{\ell}_L, \hat{\ell}_R)$, vertical force $\text{ReLU}(\hat{F}_z/\text{BW})$, and vertical acceleration envelope;
5. **Zero-Moment Point Head (`head_zmp`, $256 \to 2$):** Predicts planar $(\widehat{\text{ZMP}}_x, \widehat{\text{ZMP}}_z)$ offsets for whole-body MPC monitoring.

Because unit quaternions $\mathbf{q}$ and $-\mathbf{q}$ parameterize the exact same 3D rotation in SO(3), naive ℓ_2 regression creates artificial gradient cliffs whenever trajectories cross hemisphere boundaries during wide `ARM SWING` rotations. We train end-to-end with an **antipodal-invariant chordal quaternion objective** coupled to velocity, root, contact, and force losses:

$$\mathcal{L}_{\text{total}} = \frac{1}{B}\sum_{b=1}^{B} \min\Big(\|\hat{\mathbf{q}}_b - \mathbf{q}_b\|_2^2,\ \|\hat{\mathbf{q}}_b + \mathbf{q}_b\|_2^2\Big) + 0.35\,\mathcal{L}_{\text{vel}} + 0.25\,\mathcal{L}_{\text{root}} + 0.15\big(\mathcal{L}_{\text{BCE,contact}} + \mathcal{L}_{F_z}\big), \tag{5}$$

optimized via AdamW (lr $= 1.8\times10^{-3}$, weight decay 10^{-4}, gradient norm clip 1.0, cosine annealing over 100 epochs, batch size 64).

4.3 Within-Sequence Reconstruction, Contiguous Temporal Split, and Kinematic Rollout

Table 1 summarizes quantitative benchmarks sealed under SHA-256 receipt `d066e7f5c602212a9e4d660b3584ac6f675b6012cfd68849d7b6d6a2bb31d061` (`results/krumprobot_receipt_20261018.json`) and temporal ablation receipt `results/deep_review_ablation_receipt_20261111.json`. Across 100 epochs on a 4-vCPU Modal Cloud container (547.06 s wall-clock), multi-task loss converged to 0.00976. We explicitly separate three evaluation regimes:

- **Within-Sequence Teacher-Forced Reconstruction:** Evaluating single-step geodesic error $\theta_{\text{err}} = 2\arccos(|\langle \hat{\mathbf{q}}_j, \mathbf{q}_j \rangle|)$ across the full 2,953-frame trajectory achieves 0.0405 **rad** (2.32°) mean per-bone angular error. Single-step CPU forward inference executes in $p_{50} =$ 2.696 **ms** (2.70 **ms**) and $p_{99} = 4.069$ **ms** (4.07 **ms**), comfortably satisfying > 240 Hz real-time policy loops.

- **Zero-Overlap 80%/20% Contiguous Temporal Block Split:** Because stride-2 sliding windows across a single solo capture overlap temporally, we additionally evaluated a strict chronological split (`run_deep_review_ablations.py`) training exclusively on frames $[0, 2{,}362)$ (1,169 base windows $\rightarrow$ 3,507 mirror/noise-augmented windows) and testing with zero frame overlap on future frames $[2{,}362, 2{,}953)$ (283 held-out windows). Under this split, training geodesic error reaches 1.42° whereas unseen future freestyle windows yield 18.58° versus 2.95° for a zero-velocity persistence baseline ($\hat{\mathbf{q}}_{t+1} = \mathbf{q}_t$ at 30 fps), candidly quantifying single-sequence memorization and motivating multi-dancer training on public corpora such as AIST++ [18].

- **2,950-Step Kinematic Phase-Conditioned Rollout:** Seeded solely with the initial $W = 24$-frame standing window at $t = 0$ and driven autoregressively in kinematic WebGL space for 2,950 consecutive steps (98.37 s) using only the 140 BPM harmonic phase clock $(\sin \phi_t, \cos \phi_t)$ and six-section primitive schedule, per-bone S^3 projection prevents quaternion norm drift ($\|\hat{\mathbf{q}}_{j,t}\|_2 \equiv 1$), sustaining bounded rhythmic bounce and zero kinematic blow-up across the entire track.

5 Bipedal Kinematics, Centroidal Dynamics, and Musical Sequencing

5.1 Analytical Grounded Kinematics and Unitree UniFoLM Alignment

To deploy retargeted and neural-generated trajectories across physical humanoid morphologies aligned with the open **Unitree Robotics UniFoLM** ecosystem (`UniFoLM-WMA` world-model-action for Unitree H1 360 N · m and Unitree G1, `UniFoLM-WLA`, and `UniFoLM-VLA`) [9], our kinematic execution engine enforces exact foot-floor contact and multi-link whip physics:

1. **Grounded Two-Link Leg Inverse Kinematics (`solveKneeIK`):** Given pelvis hip joint $\mathbf{p}_{\text{hip}}$ and floor-anchored ankle target $\mathbf{p}_{\text{ankle}}$ with thigh $L_1 = 0.40$ m and shank $L_2 = 0.42$ m, the exterior anatomical knee flexion angle $\theta_{\text{flex}} = \pi - \gamma$ (0° at full extension $D = L_1 + L_2$, where γ is the interior thigh–shank angle) is solved analytically via the law of cosines:

$$D = \|\mathbf{p}_{\text{ankle}} - \mathbf{p}_{\text{hip}}\|_2, \quad \theta_{\text{flex}} = \pi - \gamma = \arccos\left(\text{clip}\left(\frac{D^2 - L_1^2 - L_2^2}{2L_1 L_2}, -1, 1\right)\right), \tag{6}$$

   enforcing strictly anterior knee flexion ($\theta_{\text{flex}} \geq 18°$, i.e., $\gamma \leq 162°$) to avoid singular straight-leg hyperextension while clamping stance-foot soles precisely to the ground plane ($Y_{\text{sole}} = -1.15$ m) with dynamic contact-shadow scaling.

2. **Whip-Lag Hierarchical Arm Forward Kinematics (`solveArmFK`):** Across a 0.42 m bi-acromial shoulder span with upper arm $L_1 = 0.28$ m, forearm $L_2 = 0.26$ m, and articulated fist geometry, distal wrist rotation trails proximal shoulder angular acceleration by a biomechanically calibrated whip delay $\Delta t_{\text{whip}} \approx 32$ ms, reproducing authentic centrifugal `ARM SWING` snap rather than rigid-rod rotation.

Table 1: Quantitative verification across extraction, neural policy learning, temporal split ablation, and centroidal dynamics.

Subsystem / Metric	**Measured Value**	**Design Target**	**Verification Receipt**
Source capture (*Fractured Cosmos*)	2,953 frames @ 29.99 fps (98.46 s)	Full 58-bar @ 140 BPM	`compare.mp4`
MPS streaming resident RAM	~ 200 MB (16-frame chunks)	< 2 GB (zero swap)	`Motion2MixamoRig`
22-bone FK rest calibration	8.765×10^{-11} error	$< 10^{-6}$	`mixamo_rotations.npz`
Skinned mesh surface error	1.42×10^{-5} cm (5.17 MB GLB)	$< 10^{-2}$ cm	`mixamo_character.glb`
`KrumpRobotPolicy` training	100 epochs, 4,392 windows, 547.06 s	Converged < 600 s	Modal Cloud CPU/PyTorch
Multi-task chordal loss $\mathcal{L}_{\text{total}}$	0.00976	< 0.020	$G_{\text{KrumpRobot}}$ PASS
Within-sequence reconstruction	0.0405 rad (2.32°)	$< 5.0°$	Full-track teacher-forced
80%/20% contiguous temporal split	1.42° train / 18.58° held-out test	2.95° persistence floor	Zero-overlap block split
Inference latency (p_{50}/p_{99})	2.70 ms / 4.07 ms	$p_{50} < 3.0$ ms	Sub-frame real-time
Kinematic phase rollout	2,950 steps ($\|\hat{\mathbf{q}}\|_2 = 1.000$)	Zero norm/pose blow-up	WebGL beat-phase driven
`STOMP` ground impact (F_{peak})	2,623.19 N ($3.82 \times$ BW, 24.5 ms)	$> 3.5 \times$ BW, < 30 ms	500 Hz 28-DoF model
`CHEST POP` arrest & peak jerk	18.2 ms arrest, 1,420 rad/s^3	< 25 ms arrest	Vitality crescendo-lock
Support-polygon ZMP excursion	4.2 cm	< 8.0 cm (no tip-over)	$G_{\text{PhysicalAI}}$ PASS
4-humanoid circle clearance	$d_{\min} = 1.28$ m (0 collisions)	> 0.80 m	16k-param onboard TRMP

5.2 500 Hz 28-DoF Analytical Centroidal Dynamics and Edge Coordination

To verify dynamic balance feasibility prior to physical hardware deployment, we evaluated a 500 Hz, 40-step-horizon analytical 28-DoF rigid-body centroidal dynamics and contact-impedance model [10] ($m = 70.0$ kg, peak actuator torque 320 N · m, `results/physical_ai_amass_papercclip_receipt_20261017.json`):

- **High-Impedance `STOMP` Transition:** Switching knee/ankle gains from compliant swing tracking to feedforward stiff impact damping ($K_p = 850$ N · m/rad, $K_d = 65$ N · m · s/rad) delivers $F_{\text{peak}} = 2{,}623.19$ N ($3.82 \times$ BW) within 24.5 ms $<$ 30.0 ms.
- **Thoracic `CHEST POP` Arrest & ZMP Margin:** Anterior sternum thrust (+0.16 m surge, +0.08 m elevation) brakes to zero velocity within 18.2 ms $<$ 25.0 ms at 1,420 rad/s^3 peak angular jerk (40 ms isometric lock), while stance-foot ZMP excursion remains bounded at 4.2 **cm** $<$ 8.0 **cm**.
- **Sub-Millisecond Multi-Robot Session Swaps:** Cloud foundation models incur prohibitive $>$ 100 ms round-trip latency for close-range multi-robot cyphers. Compiling four-humanoid pairwise spatial swap constraints onto an onboard 16k-parameter **Tiny Recursive Message Passing (TRMP)** relational graph [19] maintains minimum inter-robot clearance at $d_{\min} = 1.28$ **m** > 0.80 **m** (0 collisions) in microsecond edge cycles without external motion-capture cameras.

5.3 Full 1:38 (58-Bar @ 140 BPM) Musical Choreographic Architecture

Rather than looping a short 10-second motion clip, `KrumpRobot` synchronizes across the complete 98.46 s (58-bar) original 140 BPM soundtrack (`krump_track.mp3`) through six structured choreographic phrases (Table 2), with sub-frame spherical/linear keyframe interpolation guaranteeing tear-free 60 FPS rendering.

Table 2: Six-phrase full-track musical choreography synchronized to `krump_track.mp3` (58 bars @ 140 BPM).

Choreographic Section	Bars	Timestamp	Kinematic Primitives & Dynamic Profile
1. **Session Intro**	1–8	`0:00-0:14`	Grounded wide `BUCK` stance, 140 BPM baseline pelvis groove bounce, pre-strike torque buildup.
2. **Stomp Drop**	9–20	`0:14-0:35`	12 bars of alternating left/right heavy floor `STOMP`s every 2 bars (3.43 s), 3.82 × BW vertical spikes and radial shockwaves.
3. **Jab Combos**	21–30	`0:35-0:52`	10 bars of ballistic linear `JAB` flurries, opposing thoracic counter-rotation, and crisp forearm recoil.
4. **Power Swings**	31–42	`0:52-1:12`	12 bars of wide centrifugal `ARM SWING` whip arcs with 32 ms wrist lag and figure-8 pelvic weight transfer.
5. **Chest Climax**	43–52	`1:12-1:28`	Peak session intensity ($E = 1.25$), rapid sternum `CHEST POP`s (18.2 ms arrest), and multi-joint resonant rebounds.
6. **Kill-Off Freeze**	53–58	`1:28-1:38.5`	Decelerated tension drop locking into an asymmetrical terminal `KILL-OFF` stance held through acoustic decay.

6 Interactive WebGL Deployment, Limitations, and Sim-to-Real Roadmap

Open-Source Model Hub and Responsive 4-Mode Three.js Simulator. Trained checkpoints, JSON weights, and full-track motion tensors are openly hosted at https://huggingface.co/socialprescribing/KrumpRobot, paired with a zero-install Three.js WebGL simulator [20]

at https://huggingface.co/spaces/socialprescribing/KrumpRobot. Users can switch seamlessly at 60 FPS across four synchronized presentation modes—`Unitree H1 (Robotics)`, `Unitree G1 (Dexterous)`, `Mixamo Y-Bot (Skinned Mesh)`, and `Dual View (Side-by-Side)`—with responsive mobile touch gestures (single-finger orbital rotation, two-finger pinch zoom, double-tap camera reset) and live telemetry readouts (F_z/BW, angular jerk, ZMP, and bar/beat counter).

Limitations and Sim-to-Real Hardware Transfer Roadmap. Two honest boundaries motivate ongoing physical trials:

1. *Single-Performer Training Corpus & Cross-Performer Generalization:* Although sagittal mirror symmetry and Gaussian noise expand *Fractured Cosmos* to 4,392 windows across all five Krump primitives, our zero-overlap 80%/20% temporal block split (1.42° train vs. 18.58° held-out freestyle test) demonstrates that `KrumpRobotPolicy` memorizes the solo choreographic trajectory of a single practitioner at 140 BPM. Generalizing across unseen performers, diverse body morphologies, and variable 135–155 BPM tempos requires training and benchmarking on multi-performer 3D street-dance corpora such as AIST++ [18] paired with expanded multi-dancer Krump session captures.

2. *Kinematic WebGL Rollouts & Analytical Centroidal Model vs. Closed-Loop Whole-Body RL Control:* Our 2,950-step WebGL rollouts are kinematic pose trajectories and our 500 Hz 28-DoF centroidal dynamics check establishes analytical ZMP feasibility (4.2 cm) and impedance budgets, rather than closed-loop torque-controlled physics execution under actuator thermal derating, planetary-gearbox backlash, and compliant rubber-sole floor slip. Immediate next steps benchmark `KrumpRobotPolicy` reference motions inside massively parallel MuJoCo and NVIDIA Isaac Lab physics environments against established whole-body humanoid RL tracking controllers—specifically Expressive Whole-Body Control (*ExBody*) [21], *OmniH2O* [22], and contact-aware MPC [10]—for physical sim-to-real deployment on Unitree G1 and Unitree H1 hardware.

7 Conclusion

`KrumpRobot` demonstrates that bipedal humanoid motion generation need not sacrifice expressive vitality for kinematic stability. By uniting memory-efficient monocular 3D SMPL-X motion recovery, machine-precision rest-relative 22-bone calibration, antipodal-invariant causal GRU policy learning, and grounded bipedal kinematics across a complete 1:38 Krump performance, this work establishes an open, reproducible foundation for high-jerk expressive Physical AI.

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